

EXPERIMENT – 4

Demonstration of Hash Cracking Using John the Ripper Tool in Kali Linux

Aim

To demonstrate password hash cracking using the John the Ripper tool in Kali Linux for understanding password vulnerabilities and security analysis.

Software Requirements

Host Operating System: Windows / Linux / macOS

Virtualization Software: Oracle VirtualBox

Guest Operating System: Kali Linux

Security Tool: John the Ripper

System Requirements:

- o RAM: Minimum 4 GB (8 GB recommended)
- o Disk Space: Minimum 30 GB
- o Processor: Intel/AMD with Virtualization support

Introduction

Password hashes are commonly used to store user credentials securely. However, weak passwords and outdated hashing techniques can be exploited using password-cracking tools.

John the Ripper is a popular open-source password auditing and cracking tool used to identify weak passwords by performing dictionary, brute-force, and hybrid attacks. Using Kali Linux in a virtualized environment allows safe and ethical experimentation without affecting real systems.

Procedure

Step 1: Start Kali Linux Virtual Machine

1. Open Oracle VirtualBox.
2. Start the Kali Linux virtual machine.
3. Log in using Kali credentials.

Step 2: Open Terminal

1. Click on **Terminal** from the Kali Linux menu.
2. Ensure John the Ripper is installed:
3. `john --version`

Step 3: Prepare the Hash File

1. Create a text file containing password hashes (example: hash.txt).
2. Store the hash inside the file:
3. `nano hash.txt`
4. Save and exit the file.

Step 4: Run John the Ripper

1. Execute the John the Ripper command:
2. `john hash.txt`
3. The tool starts cracking the hash using default wordlists.

Step 5: View Cracked Passwords

1. To display cracked passwords:
2. `john --show hash.txt`

Output

```

student@kali:~$ john --version
Unknown option: "--version"

student@kali:~$ john
John the Ripper 1.9.0-jumbo-1-bleeding-aec1328d6c 2021-11-02 10:45:52 +0100 GMP [linux-gnu 64-bit x86_64 AVX2 AC]
Copyright (c) 1996-2021 by Solar Designer and others
Homepage: https://www.openwall.com/john/

Usage: john [OPTIONS] [PASSWORD-FILES]

Use --help to list all available options.

student@kali:~$ echo -e "password1\npassword2\n5f4dcc3b5aa765d61d8327deb882cf99" > hash.txt

student@kali:~$ john hash.txt
Warning: detected hash type "LM", but the string is also recognized as "dynamicmd5($p)"
Use the "--format=dynamicmd5($p)" option to force loading these as that type instead
Warning: detected hash type "LM", but the string is also recognized as "MD4A-128-4"
Use the "--format=MD4A-128-4" option to force loading these as that type instead
Warning: detected hash type "LM", but the string is also recognized as "MD2"
Use the "--format=MD2" option to force loading these as that type instead
Warning: detected hash type "LM", but the string is also recognized as "md2c"
Use the "--format=md2c" option to force loading these as that type instead
Warning: detected hash type "LM", but the string is also recognized as "mscash"
Use the "--format=mscash" option to force loading these as that type instead
Warning: detected hash type "LM", but the string is also recognized as "mscash2"
Use the "--format=mscash2" option to force loading these as that type instead
Warning: detected hash type "LM", but the string is also recognized as "NT"
Use the "--format=NT" option to force loading these as that type instead
Warning: detected hash type "LM", but the string is also recognized as "Raw-MD4"
Use the "--format=Raw-MD4" option to force loading these as that type instead
Warning: detected hash type "LM", but the string is also recognized as "Raw-MD5"
Use the "--format=Raw-MD5" option to force loading these as that type instead
Warning: detected hash type "LM", but the string is also recognized as "Raw-SHA1-AxCrypt"
Use the "--format=Raw-SHA1-AxCrypt" option to force loading these as that type instead
Warning: detected hash type "LM", but the string is also recognized as "ripemd-128"
Use the "--format=ripemd-128" option to force loading these as that type instead
Warning: detected hash type "LM", but the string is also recognized as "Snefru-128"
Use the "--format=Snefru-128" option to force loading these as that type instead

```

```
(student@kali)-[~]
$ john --format=Raw-MD5 hash.txt
Using default input encoding: UTF-8
Loaded 1 password hash (Raw-MD5 [MD5 256/256 AVX2 8x3])
Warning: no OpenMP support for this hash type, consider --fork=2
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist:/usr/share/john/password.lst
password (?)
1g 0:00:00:00 DONE 2/3 (2026-08-25 04:26) 3.125g/s 1200p/s 1200c/s 1200C/s 123456..larry
Use the "--show --format=Raw-MD5" options to display all of the cracked passwords reliably
Session completed.

(student@kali)-[~]
$ john --show --format=Raw-MD5 hash.txt
?:password
Check iptables version
1 password hash cracked, 0 left

(student@kali)-[~]
$
```

```
(student@kali)-[~]  
$ john --show hash.txt  
0 password hashes cracked, 2 left  
  
(student@kali)-[~]  
$
```

John the Ripper successfully cracks weak password hashes.

The cracked password is displayed in plaintext format.

Result

Hash cracking was successfully demonstrated using the John the Ripper tool in Kali Linux.

The experiment highlights the importance of using strong passwords and secure hashing algorithms.

Reference

<https://www.kali.org/tools/john/>

Note

This experiment is conducted **only for educational and ethical purposes** in a controlled laboratory environment.